Optimization Methods – Notes

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1 Lesson 01 — 24/09/2026

Definition 1.1 (Convex set)**.**
A set C is *convex* if, for every $x, y \in C$ and every $\alpha \in [0,1]$,

$$\alpha x + (1-\alpha)y \in C.$$

Definition 1.2 (Affine set)**.**
A set C is *affine* if, for every $x, y \in C$ and every $\alpha \in \mathbb{R}$,

$$\alpha x + (1-\alpha)y \in C.$$

Remark. Every affine set is convex, since $[0,1] \subset \mathbb{R}$.

Example of affine set are: any single point, any line, the solution set of a linear system

Definition 1.3 (convex combination)**.**
A *convex combination* of the point $x^1, x^2, \dots, x^k$ is a point

$$y = \sum_{i=1}^{k} \alpha_i x^i \quad \text{where} \quad \alpha_1, \dots, \alpha_k \in [0,1] \quad \text{and} \quad \sum_{i=1}^{k} \alpha_i = 1$$

Remark. By definition, a set $C \in R^n$ is *convex* id it contains all the convex combinations of any two points in C.

A convex set also contain any convex combination of its points.

Lemma 1.4.
if C is convex, then any $x^1, \dots, x^k \in C and \alpha_i, \dots, \alpha_k \in [0,1] s.t. \sum_{i=1}^{k} \alpha_i = 1$,

$$\sum_{i=1}^{k} \alpha_i x^i \in C$$

Definition 1.5 (convex hull)**.**
The *convex hull* $conv(C)$ of a set C is the intersection of all teh convex sets containing C.
$\implies$ it is the smallest convex set containing C

Lemma 1.6.
$conv(C) = \{all\ convex\ combinatoins\ of\ points\ in\ C\}$

Remark. Observe tha C is convex if and only $C = conv(C)$

polyhedron and balls, !ToDo

2 Lesson 02 — 25/09/2026

2.1 Example on balls

we start the lesson by giving some example of balls exercise 1.1 find the unit ball $B(0,1)$ w.r.t $\|.\|_1, \|.\|_\infty \text{and} \|.\|_A where$

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

$\|x\|_1 = |x_1| + |x_2| \quad B(1,0) = \{(x_1, x_2) : \|x\|_1 \leq 1\} \implies |x_1| + |x_2| \leq 1$

$$|x_1| + |x_2| = \begin{cases} x_1 + x_2 \\ x_1 - x_2 \\ x_2 - x_1 \\ -x_1 - x_2 \end{cases}$$

These four equations need to be all less or equal then 1 so they draw a square rotated by 45 degrees intercepting 1 or-1 on the axis, that is our Ball and it is a convex set.

ToDo ∞

Now if we consider A norm.

$\|x\|_A = \sqrt{x^t AX}$

$x^t Ax = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 4{x_1}^2 + {x_2}^2 \leq 1$ That's an ellipse

Exercise 2:
Prove that $B(0,1) = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$is convex for any norm
Let $x, y \in B(0,1)$, I have to prove that$\alpha x + (1-\alpha)y \in B(0,1), \quad \forall \alpha \in [0,1]$
$\implies \|\alpha x + (1-\alpha)y\| \leq 1 \quad \forall \alpha \in [0,1]$
$\|\alpha x + (1-\alpha)y\| \leq \|\alpha x\| + \|(1-\alpha)y\| = \alpha \|x\| + (1-\alpha)\|y\| \leq \alpha * 1 + (1-\alpha) * 1 = 1$

2.2 Operation that preserve convexity

Algebraic operations:

Lemma 2.1.
If $C_1 and C_2$ are convex, then $C_1 + C_2 \doteq \{x + y : x \in C_1, y \in C_2\}$ is convex.
If $C_1 and C_2$ are convex, then $C_1 - C_2 \doteq \{x - y : x \in C_1, y \in C_2\}$ is convex.

Topological operations:

Lemma 2.2.
if C is ocnvex then$cl(C)$is convex. if C is ocnvex then$int(C)$is convex, provided that $int(C) \neq \emptyset$.

2.3 Relative interior

Given a set $C \subseteq R^n$ we denote by $aff(C)$ teh smallest affine set containing C.

Definition 2.3 (Relative interior)**.**
Let $C \subseteq R^n$ be a convex set.
The relative interior of C is define by

$$ri(C) = \{x \in C : \exists \epsilon > 0 \ s.t. \ aff(C) \cap B(x, \epsilon) \subset C \subseteq\}$$

convex cone ToDo

3 Lesson 03 — 01/10/2026

Definition 3.1 (cone)**.**
A set C is a *cone* if, for every $x \in C$ and every $\lambda \geq 0$,

$$\lambda x \in C.$$

In other words, if C contains a point x different from 0, then it also contains the whole half-line starting at 0 and passing through x.
Cone may be convex or non-convex, let's see some examples of cones:

Example 3.2.
Here are some examples of cones:

- $\mathbb{R}^n_+$ is a convex cone.
- $\{\boldsymbol{x} \in \mathbb{R}^2 : x_1 x_2 = 0\}$ is a non-convex cone.
- Given a polyhedron $P = \{\boldsymbol{x} \in \mathbb{R}^n : A\boldsymbol{x} \leq \boldsymbol{b}\}$, the recession cone of P is the set

  $$\text{rec}(P) = \{\boldsymbol{d} \in \mathbb{R}^n : \boldsymbol{x} + \alpha \boldsymbol{d} \in P \text{ for any } \boldsymbol{x} \in P, \alpha \geq 0\}$$

  This gives us the set of directions that allow us to remain in the polyhedron. It's possible to demonstrate that this is a convex set.
- Given $\bar{x} \in cl(C) \subseteq R^n$, the set

  $$T_c(\bar{x}) = \{d \in R^n : \exists\{z_k\} \subset C, \exists\{t_k\} > 0, \lim_{k\to\infty} t_k(z_k - \bar{x}) = d\} is the tangent cone of C.$$

3.1 Convex functions

Definition 3.3.
A function f its convex , for every two points $x, y \in dom(f)$ and every $\alpha \in [0, 1]$,

$$f(\alpha y + (1-\alpha)x) \leq \alpha f(y) + (1-\alpha) f(x)$$

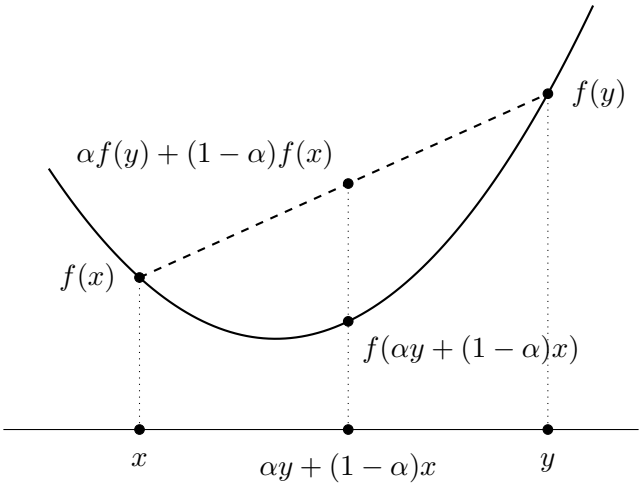


Figure 1: Example of a convex function.

Theorem 3.4.
A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if and only if the set,

$$\{(x, y) \in \mathbb{R}^n \times \mathbb{R} : y \geq f(x)\}$$

called the epigraph *of f, is convex.*

Definition 3.5 (concave function)**.**
Given a convex set $C \subseteq \mathbb{R}^n$, a function $f : R^n \to \mathbb{R}$ is concave on C if $-f$ is convex on C.

There are some functions that are both convex and concave, like a linear (affine) function $f(x) = C^t x + b$.

Theorem 3.6.
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex on the convex set $C \subseteq \mathbb{R}^n$. then f is continuous on the interior of C.

Definition 3.7 (strictly convex function)**.**
Givena a convex set $C \subseteq \mathbb{R}^n$, a function $f : \mathbb{R}^n \to \mathbb{R}$ is strictly convex on C if,

$$f(\alpha y + (1 - \alpha)x) < \alpha f(y) + (1 - \alpha)f(x)$$

for all $x, y \in C$, $x \neq y$, and $\alpha \in (0, 1)$. The difference between convex and strictly convex functions is that the inequality is strict for strictly convex functions, that means that a stictly convex function cannot be concave.

Definition 3.8 (strongly convex function)**.**
Given a convex set $C \subseteq \mathbb{R}^n$, a function $f : \mathbb{R}^n \to \mathbb{R}$ is strongly convex on C if there exists a constant $\tau > 0$ such that,

$$f(\alpha y + (1 - \alpha)x) \leq \alpha f(y) + (1 - \alpha)f(x) - \frac{\tau}{2}\alpha(1 - \alpha)\|y - x\|_2^2$$

Remark. Similarly to convex functions, we can define strictly or strongly concave functions f if $-f$ is strictly or strongly convex.

Theorem 3.9.
f is strongly convex if and only if $\exists \tau > 0$ such that $f(x) - \frac{\tau}{2}\|x\|_2^2$ is convex.

Remark. Thanks to this theorem we can say that f is strongly convex if and only if there exists a convex function g and a constant $\tau > 0$ such that $f(x) = g(x) + \frac{\tau}{2}\|x\|_2^2$.

If we consider a function $f : \mathbb{R}^n \to \mathbb{R}$ that is differentiable on an open convex set $C \subseteq \mathbb{R}^n$ we can use this theorem.

Theorem 3.10.
f is convex on C if and only if:

$$f(y) \geq f(x) + (y - x)^T \nabla f(x) \quad \forall x, y \in C$$

if we consider a function $g(y) = f(\bar{x}) + (y - \bar{x})^T \nabla f(\bar{x})$ we can see that this function is the tangent plane to the function f in the point $\bar{x}$, so this theorem says that a function is convex if and only if it is always above its tangent line.
We can also use the same theroem to ddefine strictly and strongly convex functions, we just need to change the inequality sign in the definition of convexity, for strictly convex functions, and add the term $\frac{\tau}{2}\|y - x\|_2^2$ for strongly convex functions.

Theorem 3.11.
t

- *f is strictly convex on C if and only if:*

$$f(y) > f(x) + (y-x)^T \nabla f(x) \quad \forall x, y \in C, x \neq y$$

- *f is strongly convex on C if and only if:*

$$f(y) \geq f(x) + (y-x)^T \nabla f(x) + \frac{\tau}{2}\|y-x\|_2^2 \quad \forall x, y \in C$$

If we can evaluate the second order derivative of a function $f : \mathbb{R}^n \to \mathbb{R}$ we can use the following theorem.

Theorem 3.12.
Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice differentiable function on an open convex set $C \subseteq \mathbb{R}^n$. Then:

- *f is convex on C if and only if for all $x \in C$ the Hessian matrix $\nabla^2 f(x)$ is positive semidefinite*

$$v^T \nabla^2 f(x) v \geq 0 \quad \forall v \in \mathbb{R}^n, \forall x \in C,$$

  or, equivalently, the eigenvalues of $\nabla^2 f(x)$ are non-negative for all $x \in C$.

- *if $\nabla^2 f(x)$ is positive definite for all $x \in C$, then f is strictly convex on C.*

- *f is strongly convex on C if and only if there exists a constant $\tau > 0$ such that $\nabla^2 f(x) - \tau I_n$ is positive semidefinite for all $x \in C$.*

$$v^T \nabla^2 f(x) v \geq \tau \|v\|_2^2 \quad \forall v \in \mathbb{R}^n, \forall x \in C,$$

  or, equivalently, the eigenvalues of $\nabla^2 f(x)$ are greater than or equal to τ for all $x \in C$.

4 Lesson 04 — 02/10/2026

4.1 Optimization problem in standard form

Definition 4.1 (Optimization problem)**.**
An optimization problem in `standard form` is

$$f^* = \min\{f(x) : x \in X\}, \tag{P}$$

where

- $f : \mathbb{R}^n \to \mathbb{R}$ is the `objective function`;
- $X \subseteq \mathbb{R}^n$ is the `constraints set` or `feasible region`.

If $X \equiv \mathbb{R}^n$, then (P) is said to be `unconstrained`.

From now on, we will only consider minimization problems, since

$$\max\{f(x) : x \in X\} = -\min\{-f(x) : x \in X\}.$$

4.1.1 Mathematical programming problems

In general X is defined by the expression

$$X \coloneqq \{x \in \mathbb{R}^n : g_i(x) \leq 0,\ i \in I,\ h_j(x) = 0,\ j \in J\},$$

where $g_i : \mathbb{R}^n \to \mathbb{R}$, $i \in I \coloneqq \{1, \ldots, m\}$, and $h_j : \mathbb{R}^n \to \mathbb{R}$, $j \in J \coloneqq \{1, \ldots, p\}$.

Using the notation

$$g(x) \coloneqq (g_1(x), \dots, g_m(x))^T, \qquad h(x) \coloneqq (h_1(x), \dots, h_p(x))^T,$$

we can write $X = \{x \in \mathbb{R}^n : g(x) \leq 0,\ h(x) = 0\}$, *with* $g : \mathbb{R}^n \to \mathbb{R}^m$ *and* $h : \mathbb{R}^n \to \mathbb{R}^p$. *The functions* g *and* h *are called the* ***constraint functions.***

Remark. In this case (P) is also referred to as a `mathematical programming problem`.

4.2 Global and local optima

Definition 4.2 (Optimal value)**.**
The `optimal value` of (P) is defined by

$$v(P) = \inf\{f(x) : x \in X\}.$$

- $v(P) \in \mathbb{R}$ if the problem is bounded from below;
- $v(P) = -\infty$ if the problem is unbounded from below;
- $v(P) = +\infty$ if the problem is infeasible, i.e., $X = \emptyset$.

Definition 4.3 (Global optimal solution)**.**
A `global optimal solution` of (P) is a point $x^* \in X$ such that

$$f(x^*) \leq f(x) \quad \forall x \in X.$$

The set of global minima of f on X is denoted by $X^* = \arg\min\{f(x) : x \in X\}$.

Definition 4.4 (Local optimal solution)**.**
A `local optimal solution` of (P) is a point $x^* \in X$ such that

$$f(x^*) \leq f(x) \quad \forall x \in X \cap B(x^*, r)$$

for some $r > 0$.

Example 4.5.

- $f(x) = x$, $X = \mathbb{R}$: $v(P) = -\infty$, no optimal solution.
- $f(x) = e^x$, $X = \mathbb{R}$: $v(P) = 0$, no optimal solution.
- $f(x) = x^3 - 3x$, $X = \mathbb{R}$: $v(P) = -\infty$, $x^* = 1$ is a local optimum, no global optimum.
- $f(x) = x\log(x)$, $X = \mathbb{R}_{++}$: $v(P) = -1/e$, $x^* = 1/e$ is a global optimum.
- $f(x) = 3x^4 - 8x^3 - 6x^2 + 24x + 19$, $X = \mathbb{R}$: $v(P) = 0$, $x^* = -1$ is a global optimum and $\tilde{x} = 2$ is a local optimum.

4.3 Existence of global optima

Theorem 4.6 (Weierstrass)**.**
If the objective function f *is continuous and the feasible region* X *is closed and bounded, then (at least) a global optimum exists.*

Proof. Let $v(P) = \inf_{x \in X} f(x)$. Define a minimizing sequence $\{x^k\} \subseteq X$ such that $f(x^k) \to v(P)$. Since $\{x^k\}$ is bounded, the Bolzano–Weierstrass theorem guarantees that there exists a subsequence $\{x^{k_p}\}$ converging to some point x^*. Since X is closed, we get $x^* \in X$. Finally, $f(x^{k_p}) \to f(x^*)$ since f is continuous. Therefore $f(x^*) = v(P)$, i.e., x^* is a global optimum. □

Example 4.7.
The problem

$$\min \; x_1 + x_2, \quad x \in X \coloneqq \{(x_1, x_2) : x_1^2 + x_2^2 - 4 \leq 0\}$$

admits a global optimum, since f is continuous and X is closed and bounded.

Corollary 4.8.
If the objective function f is continuous, the feasible region X is closed and there exists $k \in \mathbb{R}$ such that the k-sublevel set

$$S_k(f) = \{x \in X : f(x) \leq k\} \tag{1}$$

is nonempty and bounded, then (at least) a global optimum exists.

Proof. Minimizing f on X is equivalent to minimizing f on $S_k(f)$, which is bounded and closed since f is continuous and X is closed. Then apply the Weierstrass theorem. □

Example 4.9.
The function $f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1 - 2x_2$ fulfils condition (1). In fact, suitably choosing k, the set

$$x_1^2 + x_2^2 - 4x_1 - 2x_2 \leq k$$

is a circle with center $C = (2, 1)$ and radius $r = \sqrt{2^2 + 1^2 + k}$.

Example 4.10.
Consider

$$\min \; e^{x_1 + x_2}, \quad x \in X \coloneqq \{x_1 - x_2 \leq 0, \; -2x_1 + x_2 \leq 0\}.$$

f is continuous, X is closed and unbounded. But the sublevel set $S_2(f) = \{x \in X : f(x) \leq 2\}$ is nonempty and bounded, thus a global optimum exists. Note that $S_2(f)$ is the solution set of the system

$$\begin{cases} x_1 + x_2 \leq \log 2 \\ x_1 - x_2 \leq 0 \\ -2x_1 + x_2 \leq 0 \end{cases}$$

Corollary 4.11.
If the objective function f is continuous and `coercive`*, i.e.,*

$$\lim_{\substack{\|x\| \to \infty \\ x \in X}} f(x) = +\infty, \tag{2}$$

and the feasible region $X \neq \emptyset$ is closed, then (at least) a global optimum exists.

Proof. Let $\bar{x} \in X$ and $k \coloneqq f(\bar{x})$. By (2) the sublevel set $S_k(f)$ is nonempty and bounded, then apply the previous corollary to show that a global optimum exists. □

Example 4.12.

Consider

$$\min_{x\in\mathbb{R}} \; x^4 + 3x^3 - 5x^2 + x - 2.$$

Since f is coercive, there exists a global optimum.

Example 4.13.
The function $f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1 - 2x_2$ fulfils condition (2), with $X = \mathbb{R}^2$. Indeed,

$$f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1 - 2x_2 = (x_1 - 2)^2 - 4 + (x_2 - 1)^2 - 1.$$

If $\|x\| = \|(x_1, x_2)\| \to +\infty$, then at least one of the following conditions holds:

$$|x_1| \to +\infty \quad \text{or} \quad |x_2| \to +\infty.$$

In the first case we have

$$\lim_{\|x\|\to+\infty} f(x_1, x_2) \geq \lim_{|x_1|\to+\infty} (x_1 - 2)^2 - 5 = +\infty.$$

In the second case

$$\lim_{\|x\|\to+\infty} f(x_1, x_2) \geq \lim_{|x_2|\to+\infty} (x_2 - 1)^2 - 5 = +\infty.$$

Therefore, for any sequence $\{x^n\} = \{(x_1^n, x_2^n)\}$ such that $\|x^n\| \to +\infty$, we get

$$\lim_{n\to+\infty} f(x^n) = +\infty,$$

which guarantees that (2) is fulfilled.

5 Lesson 05 — 08/10/2026

Theorem 5.1.
Assume that f is convex on the convex set X. Then ***any local optimum of (P) is a global optimum.***

Proof. Let x^* be a local optimum of (P), i.e., there is $r > 0$ s.t.

$$f(x^*) \leq f(z) \quad \forall z \in X \cap B(x^*, r).$$

Continue on slide. □

Proposition 5.2.
Assume that f is ***strictly convex*** *on the convex set X and that (P) admits a global optimum x^*. Then x^* is the* ***unique*** *optimal solution of (P).*

Theorem 5.3.
If $f : \mathbb{R}^n \to \mathbb{R}$ is strongly convex on $\mathbb{R}^n$ and X is closed, then there exists a global optimum.

Corollary 5.4.
If f is strongly convex (on $\mathbb{R}^n$) and X is closed and convex, then there exists a unique global optimum.

Example 5.5.
Any quadratic programming problem

$$\min \frac{1}{2}x^T Qx + c^T x, \quad x \in X,$$

where Q is a `positive definite` matrix and X is closed and convex, has a unique global optimum.

If Q is positive definite, all its eigenvalues are bounded below by a strictly positive constant $\implies$ f is strongly convex.

5.1 Optimality conditions

We now focus on optimality conditions for unconstrained problems.

Theorem 5.6 (Necessary optimality condition)**.**
Assume that X is an open set and let f be differentiable at $x^* \in X$. If x^* is a local optimum of (P), then

$$\nabla f(x^*) = 0.$$

From now on, assume that f has continuous first and second order partial derivatives for every $x \in X$.

Theorem 5.7 (Second order necessary optimality condition)**.**
Let X be an open set and let $x^* \in X$ be a local optimum for (P). Then the following conditions hold:

- $\nabla f(x^*) = 0$;
- *the Hessian matrix $\nabla^2 f(x^*)$ is positive semidefinite.*

Theorem 5.8 (Second order sufficient optimality condition)**.**
Let X be an open set, $x^* \in X$, and assume that the following conditions hold:

- $\nabla f(x^*) = 0$;
- *the Hessian matrix $\nabla^2 f(x^*)$ is positive definite.*

Then x^* is a local optimum for (P).

Theorem 5.9 (Optimality condition for convex problems)**.**
Let f be a differentiable convex function on the open convex set X. Then $x^* \in X$ is a global optimum for (P) if and only if $\nabla f(x^*) = 0$.

Proof. The necessity follows from the necessary optimality condition. It's important to prove the converse implication. If $\nabla f(x^*) = 0$, we can prove it by applying the first order condition for convex functions with $y = x^*$:

$$f(x) - f(x^*) \geq (x - x^*)^T \nabla f(x^*) = 0 \quad \forall x \in X.$$

□

Theorem 5.10.
Let f be a differentiable strictly convex function on the open convex set X. Then $x^* \in X$ is

the unique global optimum for (P) if and only if $\nabla f(x^*) = 0$.

5.1.1 Existence of global optima for unconstrained quadratic programming problems

Consider the problem

$$\min f(x) := \frac{1}{2}x^T Q x + c^T x, \quad x \in \mathbb{R}^n, \tag{P}$$

where Q *is an* $n \times n$ *symmetric matrix.*

Corollary 5.11.
There exists a global optimum x^* *for (P) if and only if the following conditions hold:*

(i) $Qx^* + c = 0$*;*

(ii) Q *is positive semidefinite.*

Condition (i) is the same as $\nabla f(x^*) = 0$ *(first order condition). Condition (ii) is the second order condition (*Q *is the Hessian of* f*).*

Remark. We already observed that if Q is positive definite, then (P) admits a unique global optimum. Its expression is $x^* = -Q^{-1}c$, derived from condition (i).

If Q *is positive semidefinite but not positive definite, in order to guarantee the existence of a global optimum we need to analyze the existence of solutions of the system* $Qx + c = 0$ *using the Rouché–Capelli theorem.*

5.2 Unconstrained optimization methods

We will see some methods for optimization problems. What is an optimization methods? In an optimization descent methods we want to find a sequence of point $x^0, x^1, \dots, x^k, \dots$ *such that* $f(x^0) > f(x^1) > \dots f(x^k) > \dots$ *and we would like that* $\lim_{k\to\infty} x^k = x^*$ *where* x^* *is such that* $\nabla f(x^*) = 0$ *so it's a stationary points.*

A descent methods is characterize by the fact that $x^{k+1} = x^k + td^k$ *where* d^k *is a descent direction for* f *at* x^k *and* $t > 0$ *is the* `step`*.*

Definition 5.12 (descent direction)**.**
$d \in \mathbb{R}^n$ is a descent direction for f at x^k if we have that $d^T \nabla f(x^k) < 0$.

This means that the directional derivative along the direction d *is strictly negative. For a suitable* t *sufficiently small we have that* f *of* $f(x^{k+1}) < f(x^k)$*.*

How to find a descent direction? We need to solve $\min d^t \nabla f(x^k)$ *where the* $\|d\| = 1$*. In other words we want to find the steepest direction of descent. Recall that* $d^t \nabla f(x^k) = \|d\| \left\|f(x^k)\right\| \cos(\theta)$ *where* θ *is the angle between the gradient and the descent direction. To minimize that we need to minimize the cosine of* θ *(-1). So the solution to the min problem is* $d^k = -\frac{\nabla f(x^k)}{\|\nabla f(x^k)\|}$*. This is also known as the* `Gradient method`*.*

Algorithm 1: Gradient method

Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $\nabla f(x^k) \neq 0$ **do**
  $d^k = -\nabla f(x^k)$; // search direction
  compute an optimal solution t_k of $\min_{t>0} f(x^k + td^k)$; // step size
  $x^{k+1} = x^k + t_k d^k$;
  $k = k + 1$;

5.2.1 Gradient method

6 Lesson 06 — 09/10/2026

6.1 Gradient method — quadratic case

If $f(x) = \frac{1}{2}x^T Qx + c^T x$, with Q a positive definite matrix, we can use the Taylor expansion at x^k:

$$f(x^k + td^k) = f(x^k) + (td^k)^T \nabla f(x^k) + \frac{1}{2}(td^k)^T Q\, td^k = \frac{1}{2}(d^k)^T Q d^k\, t^2 + (d^k)^T g^k\, t + f(x^k),$$

where $g^k = \nabla f(x^k) = Qx^k + c$. Now we want to minimize this expression with respect to t: we take the derivative with respect to t, set it equal to 0 and solve for t:

$$t_k = -\frac{(d^k)^T g^k}{(d^k)^T Q d^k}.$$

As already observed, two subsequent directions are orthogonal: $(d^k)^T d^{k+1} = 0$. This implies that the generated sequence has a zig-zag behavior, which can result in slow convergence.

Theorem 6.1 (Error bound)**.**
If $f(x) = \frac{1}{2}x^T Qx + c^T x$, with Q a positive definite matrix, and x^* is the global minimum of f, then the sequence $\{x^k\}$ satisfies the following inequality:

$$\left\|x^{k+1} - x^*\right\|_Q \leq \left(\frac{\frac{\lambda_n}{\lambda_1} - 1}{\frac{\lambda_n}{\lambda_1} + 1}\right) \left\|x^k - x^*\right\|_Q \quad \forall k \geq 0, \qquad \textit{(linear convergence)}$$

where $\|x\|_Q = \sqrt{x^T Qx}$ and $0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ are the eigenvalues of Q.

Remark. If λ_n/λ_1 (the *condition number* of Q) is $\gg 1$, then the ratio

$$\frac{\frac{\lambda_n}{\lambda_1} - 1}{\frac{\lambda_n}{\lambda_1} + 1} \simeq 1$$

and the convergence may be slow.

The convergence rate is strictly related to the condition number of the matrix Q.

6.2 Inexact version of gradient methods

When f is not a quadratic function, finding the exact minimum along the direction may be computationally expensive, so in these cases we define a stopping condition on the step.

Algorithm 2: Gradient method with the Armijo inexact line search

Set $\alpha, \gamma \in (0,1)$ and $\bar{t} > 0$. Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $\nabla f(x^k) \neq 0$ **do**
  $d^k = -\nabla f(x^k)$, $t_k = \bar{t}$;
  while $f(x^k + t_k d^k) > f(x^k) + \alpha\, t_k (d^k)^T \nabla f(x^k)$ **do**
    $t_k = \gamma\, t_k$;
  $x^{k+1} = x^k + t_k d^k$;
  $k = k + 1$;

7 Lesson 07 — 15/10/2026

7.1 Conjugate gradient method

The `conjugate gradient method` *(CG) is a descent method where the search direction involves the gradient computed at the current iteration and the direction computed at the previous iteration.*

We first consider the quadratic case:

$$f(x) = \frac{1}{2} x^T Q x + c^T x,$$

where Q is positive definite. Set $g^k = \nabla f(x^k) = Qx^k + c$.

At iteration k, the search direction is defined by

$$d^k = \begin{cases} -g^0 & \text{if } k = 0, \\ -g^k + \beta_k d^{k-1} & \text{if } k \geq 1, \end{cases}$$

where β_k is such that d^k and d^{k-1} are `conjugate` *with respect to Q, i.e.,*

$$(d^k)^T Q d^{k-1} = 0.$$

From the previous condition we can compute β_k:

$$\beta_k = \frac{(g^k)^T Q d^{k-1}}{(d^{k-1})^T Q d^{k-1}}.$$

If we perform an exact line search, then d^k is a descent direction. The step size given by the exact line search is

$$t_k = -\frac{(g^k)^T d^k}{(d^k)^T Q d^k}.$$

Algorithm 3: Conjugate gradient method for quadratic functions

Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $g^k = \nabla f(x^k) \neq 0$ **do**
  if $k = 0$ **then**
    $d^k = -g^k$;
  else
    $\beta_k = \dfrac{(g^k)^T Q d^{k-1}}{(d^{k-1})^T Q d^{k-1}}$, $d^k = -g^k + \beta_k d^{k-1}$;
  $t_k = -\dfrac{(g^k)^T d^k}{(d^k)^T Q d^k}$;
  $x^{k+1} = x^k + t_k d^k$;
  $k = k + 1$;

Example 7.1.
Consider $f(x) = x_1^2 + 10x_2^2$, with starting point $x^0 = (10, 1)$.

$$Q = \begin{pmatrix} 2 & 0 \\ 0 & 20 \end{pmatrix}, \qquad \nabla f(x) = (2x_1, 20x_2).$$

- $k = 0$: $g^0 = (20, 20)$, $d^0 = -g^0 = (-20, -20)$, $t_0 = -((g^0)^T d^0)/((d^0)^T Q d^0) = 1/11$, and consequently

$$x^1 = x^0 + t_0 d^0 = (10 - 20/11, 1 - 20/11) = (90/11, -9/11).$$

- $k = 1$: $g^1 = (180/11, -180/11)$, $\beta_1 = ((g^1)^T Q d^0)/((d^0)^T Q d^0) = 81/121$, $d^1 = -g^1 + \beta_1 d^0 = (-3600/121, 360/121)$, $t_1 = -((g^1)^T d^1)/((d^1)^T Q d^1) = 11/40$, and $x^2 = x^1 + t_1 d^1 = (0, 0)$, which is the global minimum of f.

A peculiarity of this method is that for a quadratic function with positive definite Hessian it converges in at most n iterations in $\mathbb{R}^n$ (in the example, $n = 2$), as stated by the following theorem.

Theorem 7.2 (Convergence)**.**
The CG method finds the global minimum in at most n iterations. If Q has r distinct eigenvalues, then the CG method finds the global minimum in at most r iterations.

If we did not find the global minimum after k iterations, then the gradients $\{g^0, \dots, g^k\}$ are pairwise orthogonal, the directions $\{d^0, d^1, \dots, d^k\}$ are conjugate w.r.t. Q, and x^k is the minimum of f on $x^0 + \mathrm{Span}(d^0, d^1, \dots, d^{k-1})$.

Theorem 7.3 (Error bound)**.**
If $0 < \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$ are the eigenvalues of Q, then the following bounds hold:

$$\left\|x^k - x^*\right\|_Q \le 2\left(\frac{\sqrt{\lambda_n/\lambda_1} - 1}{\sqrt{\lambda_n/\lambda_1} + 1}\right)^k \left\|x^0 - x^*\right\|_Q, \qquad \forall\, k \ge 0,$$

$$\left\|x^k - x^*\right\|_Q \le \left(\frac{\lambda_{n-k+1} - \lambda_1}{\lambda_{n-k+1} + \lambda_1}\right) \left\|x^0 - x^*\right\|_Q, \qquad \forall\, k \ge 0.$$

At iteration n, notice that the error goes to 0, since $k = n$ and $\lambda_{n-k+1} - \lambda_1 = 0$.

7.2 Newton method

7.2.1 Basic version

We want to find a stationary point of a function f, i.e., a point where $\nabla f = 0$. To study the gradient, at each iteration k we consider its linear expansion at x^k:

$$\nabla f(x) \simeq \nabla f(x^k) + \nabla^2 f(x^k)(x - x^k).$$

So, to find x^{k+1}, instead of solving $\nabla f(x) = 0$ we solve the linear system

$$\nabla f(x^k) + \nabla^2 f(x^k)(x - x^k) = 0.$$

Note that x^{k+1} is a stationary point of the quadratic approximation of f at x^k:

$$f(x) \simeq f(x^k) + (x - x^k)^T \nabla f(x^k) + \frac{1}{2}(x - x^k)^T \nabla^2 f(x^k)(x - x^k).$$

Algorithm 4: Newton method (basic version)

Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $\nabla f(x^k) \neq 0$ **do**
  Let d^k be the solution of the linear system $\nabla^2 f(x^k)\, d = -\nabla f(x^k)$;
  $x^{k+1} = x^k + d^k$;
  $k = k + 1$;

Theorem 7.4 (Convergence)**.**
If x^* is a local minimum of f and $\nabla^2 f(x^*)$ is positive definite, then there exists $\delta > 0$ such that for any $x^0 \in B(x^*, \delta)$ the sequence $\{x^k\}$ converges to x^* and

$$\left\|x^{k+1} - x^*\right\| \leq C \left\|x^k - x^*\right\|^2 \qquad \forall\, k > \bar{k},$$

for some $C > 0$ and $\bar{k} > 0$ (`quadratic convergence`).

Example 7.5.
The function $f(x) = 2x_1^4 + 3x_2^4 + 2x_1^2 + 4x_2^2 + x_1x_2 - 3x_1 - 2x_2$ is strongly convex because

$$\nabla^2 f(x) = \begin{pmatrix} 24x_1^2 + 4 & 1 \\ 1 & 36x_2^2 + 8 \end{pmatrix}.$$

Starting from $x^0 = (10, 5)$, the basic Newton method gives:

k	x_1^k	x_2^k	$\left\|\nabla f(x^k)\right\|$
0	10.000000	5.000000	8189.6317378
1	6.655450	3.298838	2429.6437291
2	4.421132	2.149158	721.6330686
3	2.925965	1.361690	214.6381594
4	1.923841	0.811659	63.7752575
5	1.255001	0.428109	18.6170045
6	0.823359	0.209601	5.0058040
7	0.580141	0.171251	1.0538969
8	0.492175	0.179815	0.1022945
9	0.481639	0.180914	0.0013018
10	0.481502	0.180928	0.0000002

Drawbacks of the Newton method:

- *at each iteration we need to compute both the gradient $\nabla f(x^k)$ and the Hessian matrix $\nabla^2 f(x^k)$;*
- *`local convergence`: if x^0 is too far from the optimum x^*, then the generated sequence may not converge to x^*.*

Example 7.6.
Let $f(x) = -\frac{1}{16}x^4 + \frac{5}{8}x^2$. Then

$$f'(x) = -\frac{1}{4}x^3 + \frac{5}{4}x, \qquad f''(x) = -\frac{3}{4}x^2 + \frac{5}{4}.$$

$x^* = 0$ is a local minimum of f with $f''(x^*) = 5/4 > 0$. The sequence does not converge to

x^* if it starts from $x^0 = 1$: $x^1 = -1$, $x^2 = 1$, $x^3 = -1$, ...

7.3 Newton method with line search

If f is strongly convex, then we have global convergence because d^k is a descent direction, in fact:

$$\nabla f(x^k)^T d^k = -\nabla f(x^k)^T \left[\nabla^2 f(x^k)\right]^{-1} \nabla f(x^k) < 0.$$

Algorithm 5: Newton method with (inexact) line search

Set $\alpha, \gamma \in (0,1)$ and $\bar{t} > 0$. Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $\nabla f(x^k) \neq 0$ **do**
  Let d^k be the solution of the linear system $\nabla^2 f(x^k)\, d = -\nabla f(x^k)$;
  $t_k = \bar{t}$;
  while $f(x^k + t_k d^k) > f(x^k) + \alpha\, t_k (d^k)^T \nabla f(x^k)$ **do**
    $t_k = \gamma\, t_k$;
  $x^{k+1} = x^k + t_k d^k$;
  $k = k + 1$;

Theorem 7.7 (Convergence)**.**
If f is strongly convex, then for any starting point $x^0 \in \mathbb{R}^n$ the sequence $\{x^k\}$ converges to the global minimum of f. Moreover, if $\alpha \in (0, 1/2)$ and $\bar{t} = 1$, then the convergence is quadratic.

8 Lesson 08 — 16/10/2026

Lesson missed, I did it by myself using the slides.

So far we studied unconstrained problems, where a local minimum must be a stationary point. Now we move to problems with constraints: the condition $\nabla f(x^*) = 0$ is no longer necessary, and we need optimality conditions that also take the constraints into account.

8.1 First-order optimality conditions for constrained problems

Consider the constrained optimization problem

$$\min f(x), \quad x \in X := \{x \in \mathbb{R}^n : g_i(x) \leq 0,\ i = 1, \dots, m,\ h_j(x) = 0,\ j = 1, \dots, p\}, \tag{P}$$

where f, g_i and h_j are continuously differentiable for any i, j. We write $g = (g_1, \dots, g_m)$ and $h = (h_1, \dots, h_p)$, so the constraints are $g(x) \leq 0$ and $h(x) = 0$.

To describe the feasible set around a point x^* we use two cones: one built from the geometry of X and one built from the gradients of the constraints.

Definition 8.1 (Tangent cone)**.**
The `tangent cone` at $x^* \in X$ is

$$T_X(x^*) = \left\{ d \in \mathbb{R}^n : \exists \{z_k\} \subset X,\ \exists \{t_k\} > 0,\ z_k \to x^*,\ t_k \to 0,\ \lim_{k\to\infty} \frac{z_k - x^*}{t_k} = d \right\}.$$

So $T_X(x^*)$ contains the directions along which we can approach x^* while staying in X.

Definition 8.2 (Active constraints)**.**
The set of `active constraints` at $x^* \in X$ is

$$A(x^*) = \{i : g_i(x^*) = 0\},$$

i.e., the inequality constraints satisfied with equality at x^*.

Definition 8.3 (First-order feasible direction cone)**.**
The `first-order feasible direction cone` at $x^* \in X$ is

$$D(x^*) = \left\{ d \in \mathbb{R}^n : \begin{array}{ll} d^T \nabla g_i(x^*) \leq 0 & \forall\, i \in A(x^*), \\ d^T \nabla h_j(x^*) = 0 & \forall\, j = 1, \dots, p \end{array} \right\}.$$

$D(x^*)$ is the linearization of the feasible set at x^*: only the active constraints matter, because the inactive ones remain satisfied in a neighborhood of x^*.

8.2 Abadie constraint qualification (ACQ)

The tangent cone is the "true" local description of X, but it is hard to compute. The cone $D(x^*)$ is easy to compute, since it only needs gradients. We would like the two cones to coincide.

Definition 8.4 (Abadie constraint qualification)**.**
We say that the `Abadie constraint qualification` (ACQ) holds at a point $x^* \in X$ if

$$T_X(x^*) = D(x^*).$$

Checking ACQ directly from the definition is not easy either, so the following theorem gives some sufficient conditions that are simple to verify.

Theorem 8.5 (Sufficient conditions for ACQ)**.** *a) `Affine constraints`: if g_i and h_j are affine for all $i = 1, \dots, m$ and $j = 1, \dots, p$, then ACQ holds at any $x \in X$.*

b) `Slater condition` for convex problems: if g_i are convex for all $i = 1, \dots, m$, h_j are affine for all $j = 1, \dots, p$ and there exists $\bar{x} \in X$ s.t. $g(\bar{x}) < 0$ and $h(\bar{x}) = 0$, then ACQ holds at any $x \in X$.

c) `Linear independence` of the gradients of active constraints: if $x^* \in X$ and the vectors

$$\nabla g_i(x^*) \text{ for } i \in A(x^*), \qquad \nabla h_j(x^*) \text{ for } j = 1, \dots, p$$

are linearly independent, then ACQ holds at x^*.

8.3 Karush-Kuhn-Tucker theorem

The first step is a geometric necessary condition: at a local minimum, no direction that stays in X can be a descent direction.

Lemma 8.6.
If x^* is a local minimum of (P), then

$$d^T \nabla f(x^*) \geq 0 \qquad \forall\, d \in T_X(x^*).$$

This condition uses $T_X(x^*)$, which is hard to handle. If ACQ holds, we can replace $T_X(x^*)$ with $D(x^*)$, and this leads to an algebraic system in terms of the gradients.

Theorem 8.7 (Karush-Kuhn-Tucker)**.**
If x^* is a local minimum and ACQ holds at x^*, then there exist $\lambda^* \in \mathbb{R}^m$ and $\mu^* \in \mathbb{R}^p$ s.t. (x^*, λ^*, μ^*) satisfies the `KKT system`*:*

$$\begin{cases} \nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^p \mu_j^* \nabla h_j(x^*) = 0, \\ \lambda_i^* g_i(x^*) = 0 \quad \forall\, i = 1, \dots, m, \\ \lambda^* \geq 0, \\ g(x^*) \leq 0, \\ h(x^*) = 0. \end{cases}$$

The vectors λ^* and μ^* are called `KKT multipliers`*. The KKT system can be written in a compact form using the following function.*

Definition 8.8 (Lagrangian function)**.**
The `Lagrangian function` $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ is

$$L(x, \lambda, \mu) \coloneqq f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x).$$

Then the KKT system can be formulated as:

$$\begin{cases} \nabla_x L(x, \lambda, \mu) = 0, \\ \lambda_i g_i(x) = 0, \quad i = 1, \dots, m, \\ \lambda \geq 0,\ h(x) = 0,\ g(x) \leq 0. \end{cases}$$

Remark. Since $\lambda \geq 0$ and $g(x) \leq 0$, every term $\lambda_i g_i(x)$ is ≤ 0. So the condition $\lambda_i g_i(x) = 0$ for $i = 1, \dots, m$ is equivalent to $\lambda^T g(x) = 0$, or also $\langle \lambda, g(x) \rangle = 0$ (`complementarity condition`).

Example 8.9.
Find the minimum points of the function $f(x_1, x_2) = 2x_1 + x_2$ on the set

$$X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 4\}.$$

Note that X is compact, so from the Weierstrass theorem f admits a global maximum and a global minimum on X.
The Lagrangian function is

$$L(x_1, x_2, \lambda) = 2x_1 + x_2 + \lambda(x_1^2 + x_2^2 - 4).$$

The KKT system is given by:

$$\begin{cases} 2 + 2\lambda x_1 = 0, \\ 1 + 2\lambda x_2 = 0, \\ \lambda(x_1^2 + x_2^2 - 4) = 0, \\ x_1^2 + x_2^2 \leq 4, \\ \lambda \geq 0. \end{cases}$$

Note that for $\lambda = 0$ the system is impossible (the first equation becomes $2 = 0$). So $\lambda > 0$,

the constraint is active, and we are led to solve the system:

$$\begin{cases} x_1 = -\dfrac{1}{\lambda}, \\ x_2 = -\dfrac{1}{2\lambda}, \\ x_1^2 + x_2^2 = 4, \\ \lambda \geq 0. \end{cases}$$

Then

$$\frac{1}{\lambda^2} + \frac{1}{4\lambda^2} = 4,$$

from which

$$16\lambda^2 = 5, \qquad \lambda = \pm\frac{\sqrt{5}}{4}.$$

Since $\lambda \geq 0$, we only keep $\lambda = \frac{\sqrt{5}}{4}$:

$$x_1 = -\frac{4}{\sqrt{5}} = -\frac{4\sqrt{5}}{5}, \qquad x_2 = -\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5}.$$

The global minimum exists and, as shown in the remark below, ACQ holds at every point of X, so by definition 8.7 the global minimum must solve the KKT system. Since the KKT system has only one solution, it follows that $\bar{x} = \left(-\frac{4\sqrt{5}}{5}, -\frac{2\sqrt{5}}{5}\right)$ is the global minimum point.

Remark. In the previous example ACQ is fulfilled for every $x \in X$. Indeed, there is only the constraint $g(x) = x_1^2 + x_2^2 - 4 \leq 0$, with $\nabla g(x_1, x_2) \neq (0, 0)$ for every $x \in X$ s.t. $x_1^2 + x_2^2 - 4 = 0$. So condition c) of the sufficient conditions for ACQ holds.

What about the maximum points of f on X? It is enough to set $\lambda \leq 0$ in the KKT system. In fact, to find the maxima of f we look for the minima of $-f$. The KKT system for the problem

$$\min\left(-f(x)\right), \quad x \in X$$

is

$$\begin{cases} -2 + 2\lambda x_1 = 0, \\ -1 + 2\lambda x_2 = 0, \\ \lambda(x_1^2 + x_2^2 - 4) = 0, \\ x_1^2 + x_2^2 \leq 4, \\ \lambda \geq 0, \end{cases}$$

which is equivalent to setting $\lambda \leq 0$ in the original one. Choosing $\lambda = -\frac{\sqrt{5}}{4}$ in the original system we obtain

$$x_1 = \frac{4}{\sqrt{5}} = \frac{4\sqrt{5}}{5}, \qquad x_2 = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5},$$

so that $\hat{x} = \left(\frac{4\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$ is the global maximum point of f on X.

8.3.1 Limits of the KKT theorem

The KKT theorem has two limits: it needs ACQ, and it only gives necessary conditions.

Remark. The ACQ assumption is crucial in the KKT theorem.

Example 8.10.
Consider

$$\begin{cases} \min\ x_1 + x_2 \\ (x_1 - 1)^2 + (x_2 - 1)^2 - 1 \leq 0 \\ x_2 \leq 0 \end{cases}$$

The feasible set is the single point $x^* = (1, 0)$, which is therefore the global optimum.
$T_X(x^*) = \{0\}$ and $D(x^*) = \{d \in \mathbb{R}^2 : d_2 = 0\}$, hence ACQ does not hold at x^*.
$\nabla g_1(x^*) = (0, -2)$, $\nabla g_2(x^*) = (0, 1)$, $\nabla f(x^*) = (1, 1)$, hence there is no λ^* s.t. (x^*, λ^*) solves the KKT system: the first component of $\nabla_x L$ is always $1 \neq 0$.

The KKT theorem gives necessary optimality conditions, but not sufficient ones.

Example 8.11.
Consider

$$\begin{cases} \min\ x_1^2 - x_2^2 \\ x_1^2 + x_2^2 - 2 \leq 0 \end{cases}$$

$x^* = (0, 0)$, $\lambda^* = 0$ solves the KKT system, but x^* is not a local optimum.

For convex problems, instead, the KKT conditions are also sufficient.

Theorem 8.12.
If the optimization problem is convex and (x^*, λ^*, μ^*) solves the KKT system, then x^* is a global optimum.

Recall that (P) is convex if f and g are convex and h is affine.
Exercise 4.1. Prove definition 8.12.

9 Lesson 09 — 22/10/2026

9.1 Lagrangian duality

9.1.1 Lagrangian relaxation and dual function

The Lagrangian function is also the starting point of duality theory. The idea is to remove the constraints and move them into the objective function, weighted by the multipliers. Denote by $v(P)$ the optimal value of (P).

Definition 9.1 (Lagrangian relaxation)**.**
Given $\lambda \geq 0$ and $\mu \in \mathbb{R}^p$, the problem

$$\inf L(x, \lambda, \mu), \quad x \in \mathbb{R}^n$$

is called `Lagrangian relaxation` of (P), and

$$\varphi(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \mu)$$

is the `Lagrangian dual function`.

The Lagrangian relaxation provides a lower bound to $v(P)$.

Theorem 9.2.
For any $\lambda \geq 0$ and $\mu \in \mathbb{R}^p$, we have $\varphi(\lambda, \mu) \leq v(P)$.

Proof. If $x \in X$, i.e., $g(x) \leq 0$ and $h(x) = 0$, then

$$L(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{j=1}^{p} \mu_j h_j(x) \leq f(x),$$

hence

$$\varphi(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \mu) \leq \inf_{x \in X} L(x, \lambda, \mu) \leq \inf_{x \in X} f(x) = v(P). \qquad \square$$

The dual function φ has the following properties:

- *it is concave, because it is the infimum of affine functions w.r.t. (λ, μ);*
- *it may be equal to $-\infty$ at some point;*
- *it may be not differentiable at some point.*

9.1.2 Lagrangian dual problem

Each choice of (λ, μ) gives a lower bound, so it is natural to look for the best one.

Definition 9.3 (Lagrangian dual problem)**.**
The problem

$$\max \varphi(\lambda, \mu), \quad \lambda \geq 0, \ \mu \in \mathbb{R}^p \tag{D}$$

is called `Lagrangian dual problem` of (P), and (P) is called `primal problem`.

- *The dual problem (D) consists in finding the best lower bound of $v(P)$.*
- *(D) is always equivalent to a convex problem, even if (P) is non-convex: indeed, it is the maximization of a concave function on a convex set.*

Definition 9.2 can be equivalently stated as follows.

Theorem 9.4 (Weak duality)**.**
For any optimization problem (P), we have $v(D) \leq v(P)$.

The previous inequality is called ***`weak duality property`****.*

Example 9.5 (Linear programming)**.**
Primal problem:

$$\begin{cases} \min c^T x \\ Ax \geq b \end{cases} \tag{P}$$

Lagrangian function: $L(x, \lambda) = c^T x + \lambda^T (b - Ax) = \lambda^T b + (c^T - \lambda^T A)x$.
Dual function:

$$\varphi(\lambda) = \inf_{x \in \mathbb{R}^n} L(x, \lambda) = \begin{cases} -\infty & \text{if } c^T - \lambda^T A \neq 0, \\ \lambda^T b & \text{if } c^T - \lambda^T A = 0. \end{cases}$$

Dual problem:

$$\begin{cases} \max \varphi(\lambda) \\ \lambda \geq 0 \end{cases} \longrightarrow \begin{cases} \max \lambda^T b \\ \lambda^T A = c^T \\ \lambda \geq 0 \end{cases} \tag{D}$$

which is a linear programming problem.

Exercise 4.2. Find the dual of (D).

Example 9.6 (Least norm solution of linear equations)**.**
Primal problem:

$$\begin{cases} \min \ \frac{1}{2}x^T x \\ Ax = b \end{cases} \tag{P}$$

Lagrangian function: $L(x, \mu) = \frac{1}{2}x^T x + \mu^T(b - Ax)$.
Dual function: $\varphi(\mu) = \inf_{x\in\mathbb{R}^n} L(x, \mu)$.
$L(x, \mu)$ is quadratic and strongly convex with respect to x, therefore

$$\varphi(\mu) = \inf_{x\in\mathbb{R}^n} L(x, \mu) = \min_{x\in\mathbb{R}^n} L(x, \mu),$$

thus the global optimum is the stationary point:

$$\nabla_x L = x - A^T\mu = 0 \iff x = A^T\mu,$$

hence $\varphi(\mu) = -\frac{1}{2}\mu^T AA^T\mu + b^T\mu$.
Dual problem:

$$\begin{cases} \max \ -\frac{1}{2}\mu^T AA^T\mu + b^T\mu \\ \mu \in \mathbb{R}^p \end{cases} \tag{D}$$

which is an unconstrained convex quadratic programming problem.

Exercise 4.3. Find the dual problem of a general convex quadratic programming problem

$$\begin{cases} \min \ \frac{1}{2}x^T Qx + c^T x \\ Ax \le b \end{cases} \tag{P}$$

where Q is a symmetric positive definite matrix.

9.1.3 Strong duality

Weak duality says that $v(D) \le v(P)$. The interesting case is when the two values coincide.

Definition 9.7 (Strong duality)**.**
We say that `strong duality` holds for (P) if $v(D) = v(P)$ and (D) admits an optimal solution.

Strong duality does not hold in general.

Example 9.8.
Consider the following (non-convex) problem:

$$\begin{cases} \min \ -x^2 \\ x - 1 \le 0 \\ -x \le 0 \end{cases} \tag{P}$$

It is easy to check that $v(P) = -1$. The Lagrangian function is $L(x, \lambda) = -x^2 + \lambda_1(x - 1) - \lambda_2 x$, hence

$$\varphi(\lambda) = \inf_{x\in\mathbb{R}} L(x, \lambda) = -\infty \qquad \forall\, (\lambda_1, \lambda_2) \in \mathbb{R}^2,$$

so $v(D) = -\infty$.

The next theorem provides sufficient conditions which guarantee strong duality for (P).

Theorem 9.9.
Suppose f, g, h are continuously differentiable, the primal problem

$$\begin{cases} \min\ f(x) \\ g(x) \leq 0 \\ h(x) = 0 \end{cases} \tag{P}$$

is convex, there exists a global optimum x^* and ACQ holds at x^*. Then:

- *strong duality holds ($v(D) = v(P)$ and (D) admits an optimal solution);*
- *(λ^*, μ^*) is optimal for (D) if and only if (λ^*, μ^*) is a KKT multipliers vector associated with x^*.*

Proof. $L(x, \lambda, \mu)$ is convex with respect to x since (P) is convex. Since ACQ holds at x^*, by definition 8.7 there exists a KKT vector of multipliers associated with x^*. Let (λ^*, μ^*) be any such vector. Then

$$\nabla_x L(x^*, \lambda^*, \mu^*) = 0, \qquad \lambda^* \geq 0, \qquad (\lambda^*)^T g(x^*) = 0.$$

Thus, using that L is convex in x (so a stationary point is a global minimum) and weak duality,

$$\begin{aligned} v(D) \geq \varphi(\lambda^*, \mu^*) &= \inf_x L(x, \lambda^*, \mu^*) = L(x^*, \lambda^*, \mu^*) \\ &= f(x^*) + (\lambda^*)^T g(x^*) + (\mu^*)^T h(x^*) = f(x^*) = v(P) \geq v(D). \end{aligned}$$

Therefore $v(P) = v(D)$ and (λ^*, μ^*) is optimal for (D).

Vice versa, if (λ^*, μ^*) is any optimal solution for (D), then

$$\begin{aligned} f(x^*) = v(P) = v(D) = \varphi(\lambda^*, \mu^*) &= \inf_x L(x, \lambda^*, \mu^*) \leq L(x^*, \lambda^*, \mu^*) \\ &= f(x^*) + (\lambda^*)^T g(x^*) + (\mu^*)^T h(x^*) \\ &= f(x^*) + (\lambda^*)^T g(x^*) \leq f(x^*), \end{aligned}$$

thus $(\lambda^*)^T g(x^*) = 0$ and $\inf_x L(x, \lambda^*, \mu^*) = L(x^*, \lambda^*, \mu^*)$, hence $\nabla_x L(x^*, \lambda^*, \mu^*) = 0$, i.e., (λ^*, μ^*) is a KKT multipliers vector associated with x^*. □

Convexity is sufficient but not necessary: strong duality may hold also for some non-convex problems.

Example 9.10.
Consider the (non-convex) problem

$$\begin{cases} \min\ -x_1^2 - x_2^2 \\ x_1^2 + x_2^2 - 1 \leq 0 \end{cases}$$

We have $v(P) = -1$. The Lagrangian function is

$$L(x, \lambda) = -x_1^2 - x_2^2 + \lambda(x_1^2 + x_2^2 - 1) = (\lambda - 1)x_1^2 + (\lambda - 1)x_2^2 - \lambda.$$

The dual function is

$$\varphi(\lambda) = \inf_{x \in \mathbb{R}^2} L(x, \lambda) = \begin{cases} -\infty & \text{if } \lambda < 1, \\ -\lambda & \text{if } \lambda \geq 1. \end{cases}$$

The dual problem is

$$\begin{cases} \max\ -\lambda \\ \lambda \geq 1 \end{cases}$$

hence its optimal solution is $\lambda^* = 1$ and $v(D) = -1 = v(P)$.

Finally, strong duality can be characterized through the saddle points of the Lagrangian function.

Theorem 9.11 (Characterization of strong duality)**.**
(x^*, λ^*, μ^*) *is a* ***saddle point*** *of* L*, i.e.,*

$$L(x^*, \lambda, \mu) \leq L(x^*, \lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*) \qquad \forall\, x \in \mathbb{R}^n,\ \lambda \in \mathbb{R}^m_+,\ \mu \in \mathbb{R}^p,$$

if and only if x^* *is optimal for (P),* (λ^*, μ^*) *is optimal for (D) and* $v(P) = v(D)$*.*

Proof. If (x^*, λ^*, μ^*) is a saddle point of L, then we can prove that $x^* \in X$ and $\langle \lambda^*, g(x^*) \rangle = 0$, which implies $\varphi(\lambda^*, \mu^*) = f(x^*)$.

Vice versa, we have that

$$f(x^*) = \varphi(\lambda^*, \mu^*) = \inf_{x \in \mathbb{R}^n} L(x, \lambda^*, \mu^*) \leq L(x^*, \lambda^*, \mu^*) = f(x^*) + \langle \lambda^*, g(x^*) \rangle,$$

hence $\langle \lambda^*, g(x^*) \rangle = 0$ and

$$L(x^*, \lambda^*, \mu^*) = f(x^*) = \varphi(\lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*) \qquad \forall\, x \in \mathbb{R}^n.$$

Moreover

$$L(x^*, \lambda, \mu) = f(x^*) + \langle \lambda, g(x^*) \rangle \leq f(x^*) = L(x^*, \lambda^*, \mu^*)$$

for all $\lambda \in \mathbb{R}^m_+$ and $\mu \in \mathbb{R}^p$. □